Contents

EEE 554 — Lecture #7

Discrete-Type Random Variables

Suppose x is a random variable (RV) that takes values in a discrete set

$$R = \{r_k,\ k \in \mathbb{Z}_+\}.$$

Then the CDF F_x of x is a **piecewise-constant** function:

$$F_x(x) = \sum_{r_k \le x} \Pr\{x = r_k\}.$$

- F_x has **jump discontinuities** at $x = r_k$.

“Formal derivative” using Dirac delta

Such a CDF has a formal derivative in terms of Dirac delta symbols:

$$f_x(x) = \sum_{r_k} \Pr\{x = r_k\}\, \delta(x - r_k).$$

Note:

$$\int_{-\infty}^{x} f_x(u)\, du = \int_{-\infty}^{x} \sum_{r_k} \Pr\{x = r_k\}\delta(u - r_k)\, du = \sum_{r_k \le x} \Pr\{x = r_k\} = F_x(x).$$

Also,

$$\int_{-\infty}^{\infty} f_x(x)\, dx = \sum_{r_k} \Pr\{x = r_k\} = 1.$$

Example: x = outcome of a die roll

$$R = \{1, 2, 3, 4, 5, 6\}, \qquad r_k = k,\ k = 1, \dots, 6$$

$$\Pr\{x = r_k\} = \frac{1}{6}, \qquad k = 1, \dots, 6$$

Expected value via "delta-PDF"

$$\mathbb{E}[x] = \int_{-\infty}^{\infty} x f_x(x)\, dx = \int_{-\infty}^{\infty} x \sum_{k=1}^{6} \frac{1}{6}\delta(x-k)\, dx = \sum_{k=1}^{6} \frac{1}{6}k = \frac{21}{6} = \frac{7}{2}.$$

PMF (probability mass function)

If x takes integer values, the discrete function

$$p(k) = \Pr\{x = k\}$$

is known as the **probability mass function (PMF)**, and it is equivalent to the "PDF with deltas" given above.

Using PMF:

$$\mathbb{E}[x] = \sum_{k=-\infty}^{\infty} k\, p(k) = \frac{7}{2}.$$

Mixed-Type RVs

In certain situations, the "PDF" of an RV may contain **both delta symbols and ordinary (continuous) values**. Such an RV is said to be of **mixed type**.

Example

Consider a random variable x with a standard normal distribution, and a new random variable y defined by a transformation using the unit step function $u(x)$:

$$x \sim \mathcal{N}(0, 1), \qquad y = x\, u(x)$$

Analysis of the Transformation:

- **Case 1:** $x < 0$
  - The unit step function $u(x) = 0$.
  - Therefore, $y = x \cdot 0 = 0$.
  - **Result:** All negative values of x are mapped to the single point $y = 0$.
- **Case 2:** $x \geq 0$
  - The unit step function $u(x) = 1$.
  - Therefore, $y = x \cdot 1 = x$.
  - **Result:** All positive values of x are mapped to y identically ($y = x$).

The Distribution of y:

The resulting random variable y is of **mixed type** because its distribution contains two distinct components:

1. **Discrete Component (Point Mass):**

- Since all $x < 0$ map to $y = 0$, the probability of $y = 0$ is equal to the total probability mass of x in the negative region.
- For a standard normal distribution, due to symmetry, $\Pr\{x < 0\} = 0.5$.
- This creates a **Dirac delta function** (impulse) at $y = 0$ with area (weight) 0.5.
- Mathematical Representation: $0.5\,\delta(y)$.

2. **Continuous Component:**
   - For $y > 0$, y follows the same distribution as x (the positive half of the Gaussian curve).
   - Mathematical Representation: $\frac{1}{\sqrt{2\pi}} e^{-y^2/2}$ for $y > 0$.

Probability Density Function (PDF) of y:

Combining these, the PDF of y is:

$$f_y(y) = \underbrace{0.5\,\delta(y)}_{\text{Discrete Part}} + \underbrace{\frac{1}{\sqrt{2\pi}} e^{-y^2/2}\, u(y)}_{\text{Continuous Part}}$$

Note: The Dirac delta $\delta(y)$ captures the finite probability ($\Pr\{x < 0\} = 0.5$) concentrated at the single point $y = 0$.

Multiple RVs

Joint CDF

The joint CDF of RVs $x_1, \dots, x_n$ is

$$F_{\mathbf{X}}(x_1, \dots, x_n) = \Pr\left(\{X_1 \le x_1\} \cap \cdots \cap \{X_n \le x_n\}\right).$$

The subscript $\mathbf{X}$ represents the vector containing the X_k as components:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix}.$$

Properties:

- $0 \le F_{\mathbf{X}}(x_1, \dots, x_n) \le 1$.
- If $\Delta > 0$, then increasing any coordinate cannot decrease the CDF:

$$F_{\mathbf{X}}(x_1, \dots, x_{k-1}, x_k + \Delta, x_{k+1}, \dots, x_n) \ge F_{\mathbf{X}}(x_1, \dots, x_n).$$

- Limits:

$$\lim_{\substack{x_1 \to -\infty \\ \vdots \\ x_n \to -\infty}} F_{\mathbf{X}}(x_1, \dots, x_n) = 0, \qquad \lim_{\substack{x_1 \to +\infty \\ \vdots \\ x_n \to +\infty}} F_{\mathbf{X}}(x_1, \dots, x_n) = 1.$$

Joint PDF

If $F_{\mathbf{X}}$ is differentiable, the joint PDF is

$$f_{\mathbf{X}}(x_1, \dots, x_n) = \frac{\partial^n}{\partial x_1 \cdots \partial x_n} F_{\mathbf{X}}(x_1, \dots, x_n).$$

And

$$F_{\mathbf{X}}(x_1, \ldots, x_n) = \int_{-\infty}^{x_n} \cdots \int_{-\infty}^{x_1} f_{\mathbf{X}}(u_1, \ldots, u_n)\, du_1 \cdots du_n.$$

Normalization:

$$\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{\mathbf{X}}(x_1, \ldots, x_n)\, dx_1 \cdots dx_n = 1.$$

Marginal PDFs (Marginalization)

The CDF of X_1 can be obtained from the joint CDF by allowing $x_2, \ldots, x_n$ to take on any possible value:

$$F_{X_1}(x_1) = \lim_{x_2 \to \infty, \ldots, x_n \to \infty} F_{\mathbf{X}}(x_1, \ldots, x_n).$$

The marginal PDF is obtained by integrating out ("marginalizing") the undesired variables:

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{\mathbf{X}}(x_1, x_2, \ldots, x_n)\, dx_2 \cdots dx_n.$$

Similarly, the joint PDF of any subset of the X_k can be obtained by integrating the full joint PDF over the variables you don't want. This process is called **marginalization**.

Example (marginalization in 2D)

Given a joint PDF

$$f_X(x_1, x_2) = \begin{cases} 1, & 0 \le x_1 \le 2x_2,\ 0 \le x_2 \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Compute the marginal $f_{X_2}(x_2)$:

$$f_{X_2}(x_2) = \int_{-\infty}^{\infty} f_X(x_1, x_2)\, dx_1 = \int_0^{2x_2} 1\, dx_1 = 2x_2,$$

for $0 \le x_2 \le 1$, and 0 otherwise.

From this, it is straightforward to calculate $\mathbb{E}[X_2]$, $\mathrm{Var}(X_2)$, etc.